

$$\frac{d\vec{p}_{кин}}{dt} = e\vec{E} + \frac{e}{c} [\vec{v}, \vec{H}]$$

$$\frac{d\varepsilon_{кин}}{dt} = e(\vec{E}, \vec{v})$$

$$\varepsilon_{кин} = mc^2 \gamma, \quad p_{кин} = m\vec{v}\gamma$$

$$mc \frac{du_\mu}{ds} = \frac{e}{c} F_{\mu\nu} u^\nu$$

$$F_{\mu\nu} = \frac{\partial A_\nu}{\partial x^\mu} - \frac{\partial A_\mu}{\partial x^\nu}$$

$$F_{\mu\nu} = -F_{\nu\mu}$$

$$F_{\mu\nu} = \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & -H_z & H_y \\ -E_y & H_z & 0 & -H_x \\ -E_z & -H_y & H_x & 0 \end{pmatrix} = \begin{pmatrix} 0 & \vec{E} \\ -\vec{E} & -\varepsilon_{ijk} H_k \end{pmatrix} = (\vec{E}, \vec{H})$$

1. Об-ва тензора EM поля

$$F^{\mu\nu} = g^{\mu\mu'} g^{\nu\nu'} F_{\mu'\nu'}$$

$$F^{\mu\nu} = g F_{\mu'\nu'} g^T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & -1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -H_z & H_y \\ E_y & H_z & 0 & -H_x \\ E_z & -H_y & H_x & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\vec{E} \\ \vec{E} & -\varepsilon_{ijk} H_k \end{pmatrix} = (-\vec{E}, \vec{H})$$

$$\tilde{F}^{\mu\nu} = \frac{1}{2} \varepsilon^{\mu\nu\alpha\beta} F_{\alpha\beta}$$

$\tilde{F}$ - антисимм. тензор

тензор Леви-Чивиты

$$\mu=0, \quad \nu=(x,y,z) \quad \left\{ \begin{aligned} \tilde{F}^{0\nu} &= \frac{1}{2} \varepsilon^{0\nu\alpha\beta} F_{\alpha\beta} = \frac{1}{2} \varepsilon_{\nu\alpha\beta} (-\varepsilon_{\alpha\beta\kappa} H_\kappa) = -\frac{1}{2} \varepsilon_{\alpha\beta\nu} \varepsilon_{\alpha\beta\kappa} H_\kappa = -\delta_{\nu\kappa} H_\kappa = -H_\nu \end{aligned} \right.$$

$$\mu=(x,y,z), \quad \nu=(x,y,z) \quad \left\{ \begin{aligned} \tilde{F}^{\mu\nu} &= \frac{1}{2} \varepsilon^{\mu\nu\alpha\beta} F_{\alpha\beta} + \frac{1}{2} \varepsilon^{\mu\nu\alpha 0} F_{\alpha 0} = \frac{1}{2} (\varepsilon_{\mu\nu\beta} E_\beta + \varepsilon_{\mu\nu\alpha} E_\alpha) = \varepsilon_{\mu\nu\alpha} E_\alpha \end{aligned} \right.$$

$$\tilde{F}^{\mu\nu} = \begin{pmatrix} 0 & -H_x & -H_y & -H_z \\ H_x & 0 & E_z & -E_y \\ H_y & -E_z & 0 & E_x \\ H_z & E_y & -E_x & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\vec{H} \\ \vec{H} & \varepsilon_{ijk} E_k \end{pmatrix} = (-\vec{H}, -\vec{E})$$

$$\tilde{F}_{\mu\nu} = \begin{pmatrix} 0 & H_x & H_y & H_z \\ -H_x & 0 & E_z & -E_y \\ -H_y & -E_z & 0 & E_x \\ -H_z & E_y & -E_x & 0 \end{pmatrix} = \begin{pmatrix} 0 & \vec{H} \\ -\vec{H} & \varepsilon_{ijk} E_k \end{pmatrix} = (\vec{H}, -\vec{E})$$

$$F \rightarrow \tilde{F}, \quad \vec{E} \rightarrow \vec{H}, \quad \vec{H} \rightarrow -\vec{E}$$

2. Преобразование полей $\vec{E}$ и $\vec{H}$ при переходе в новую СО

$$F'^{\mu'\nu'} = \frac{\partial x'^{\mu'}}{\partial x^\mu} \frac{\partial x'^{\nu'}}{\partial x^\nu} F^{\mu\nu} = \Lambda F^{\mu\nu} \Lambda^T \quad \text{⊕}$$

матр. преобр. Лоренца

$$\begin{pmatrix} 0 & -E'_x & -E'_y & -E'_z \\ E'_x & 0 & -H'_z & H'_y \\ E'_y & H'_z & 0 & -H'_x \\ E'_z & -H'_y & H'_x & 0 \end{pmatrix}$$

$$\text{⊕} \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -H_z & H_y \\ E_y & H_z & 0 & -H_x \\ E_z & -H_y & H_x & 0 \end{pmatrix} \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \text{⊕}$$

$$\Rightarrow \begin{pmatrix} -\beta\gamma E_x & -\gamma E_x & -\gamma E_y + \beta\gamma H_z & -\gamma E_z - \beta\gamma H_y \\ \gamma E_x & \beta\gamma E_x & \beta\gamma E_y - \gamma H_z & \beta\gamma E_z + \gamma H_y \\ E_y & H_z & 0 & -H_x \end{pmatrix} \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$E'_x = \gamma^2 E_x - \beta^2 \gamma^2 E_x = E_x$$

$$E'_y = \gamma E_y - \beta\gamma H_z$$

$$E'_z = \gamma E_z + \beta\gamma H_y$$

$$\bar{E}'_{||} = \bar{E}_{||}$$

$$\bar{E}'_{\perp} = \gamma (\bar{E}_{\perp} + \frac{1}{c} [\vec{v}, \vec{H}])$$

$$H'_x = H_x$$

$$H'_y = \gamma H_y + \beta\gamma E_z$$

$$H'_z = \gamma H_z - \beta\gamma E_y$$

$$\bar{H}'_{||} = \bar{H}_{||}$$

$$\bar{H}'_{\perp} = \gamma (\bar{H}_{\perp} - \frac{1}{c} [\vec{v}, \vec{E}])$$

3. Инварианты ЭМ поля

$$I_1 = F^{\mu\nu} F_{\mu\nu} - \text{скаляр}$$

$$I_2 = \tilde{F}^{\mu\nu} F_{\mu\nu} - \text{псевдоскаляр}$$

$$I_3 = \tilde{F}^{\mu\nu} \tilde{F}_{\mu\nu} \sim -I_1$$

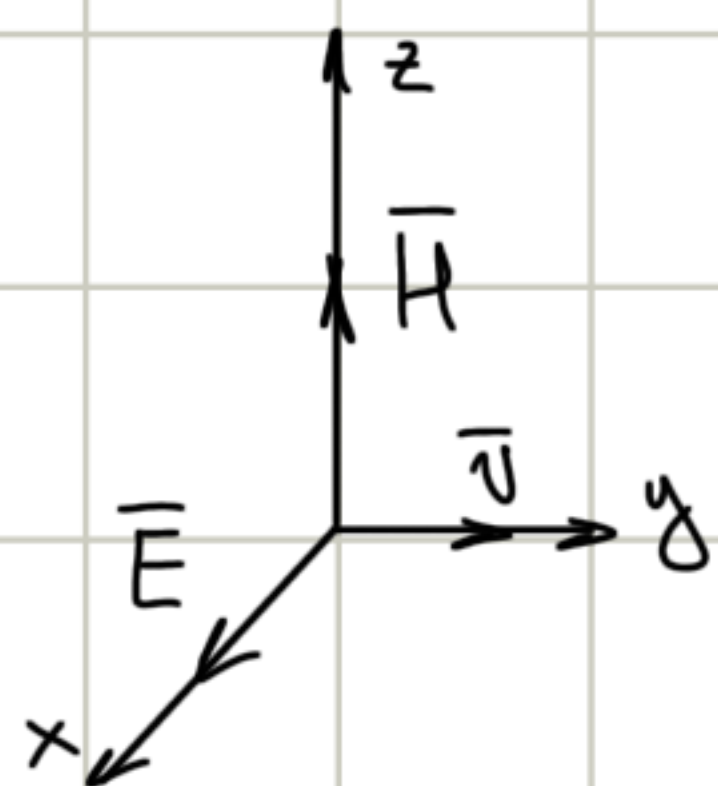
$$I_1 = F^{\mu\nu} F_{\mu\nu} = \text{tr}(F^{\mu\nu} (F_{\mu\nu})^T) = \text{tr} \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & -H_z & H_y \\ -E_y & H_z & 0 & -H_x \\ -E_z & -H_y & H_x & 0 \end{pmatrix} \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & H_z & -H_y \\ E_y & -H_z & 0 & H_x \\ E_z & H_y & -H_x & 0 \end{pmatrix} =$$

$$= -\bar{E}^2 - E_x^2 + H_z^2 + H_y^2 - E_y^2 + H_z^2 + H_x^2 - E_z^2 + H_x^2 + H_y^2 = 2(\bar{H}^2 - \bar{E}^2)$$

скаляр
(не меняет знак
при замене $\vec{r} \rightarrow -\vec{r}$)

$$I_2 = \text{tr}(\tilde{F}^{\mu\nu} (F_{\mu\nu})^T) = \text{tr} \begin{pmatrix} 0 & -H_x & -H_y & -H_z \\ H_x & 0 & E_z & -E_y \\ H_y & -E_z & 0 & E_x \\ H_z & E_y & -E_x & 0 \end{pmatrix} \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & H_z & -H_y \\ E_y & -H_z & 0 & H_x \\ E_z & H_y & -H_x & 0 \end{pmatrix} = -4(\vec{E}, \vec{H})$$

псевдоскаляр
(меняет знак
при замене $\vec{r} \rightarrow -\vec{r}$)



$$\vec{E} \perp \vec{H}$$

$$|\vec{E}| > |\vec{H}|$$

$$\Rightarrow \exists \text{ CO, где } \vec{H}' = 0, \text{ есть только поле } \vec{E}'$$

$$\vec{H}'_{\perp} = \gamma (H_{\perp} - \frac{1}{c} [\vec{v}, \vec{E}])$$

$$\Rightarrow \vec{v} = \frac{c [\vec{E}, \vec{H}]}{E^2} < c$$

переход к
такой СО

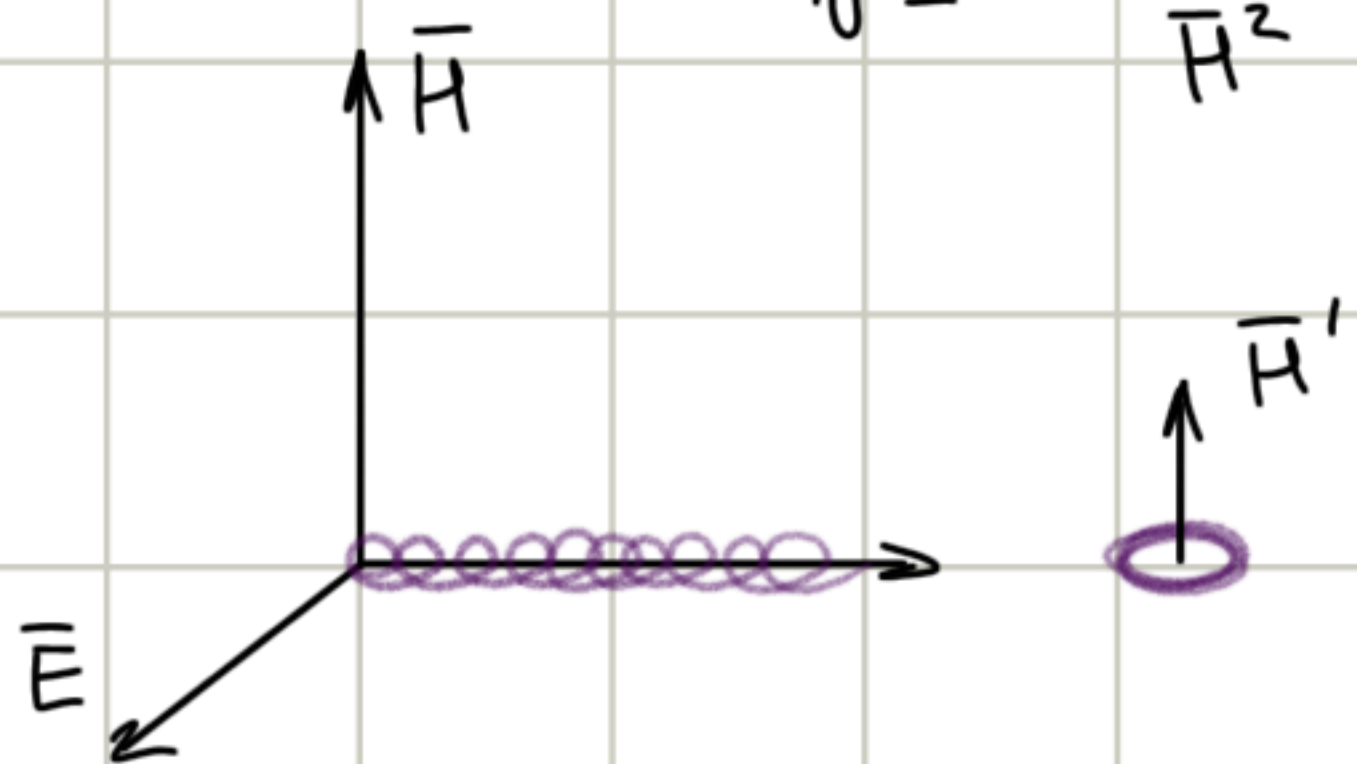
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$$\vec{v} = \frac{c [\vec{E}, \vec{H}]}{H^2} < c$$

переход к
такой СО



4 Канон. инвариантность, канон. преобраз., канонизация Лоренца и Кулона

$$A^\mu = (\varphi, \bar{A})$$

$$\bar{E} = -\frac{\partial \varphi}{\partial r} - \frac{1}{c} \frac{\partial \bar{A}}{\partial t}$$

$$F_{\mu\nu} = \frac{\partial A_\nu}{\partial x^\mu} - \frac{\partial A_\mu}{\partial x^\nu}$$

$$\bar{H} = \text{rot} \bar{A}$$

$$\bar{A}' = \bar{A} + \frac{\partial f}{\partial \bar{r}}, \quad f - \text{скаляр}$$

$$\varphi' = \varphi - \frac{1}{c} \frac{\partial f}{\partial t}$$

$$A'_\mu = A_\mu + \frac{\partial f}{\partial x^\mu}$$

каноническое преобраз., не меняет $\bar{E}$ и $\bar{H}$

Канонизация Лоренца: $\frac{\partial A^\mu}{\partial x^\mu} = 0$, $\frac{\partial \varphi}{\partial t} + \text{div} \bar{A} = 0$ явл. Лоренц. инвар.

Канонизация Кулона: $\text{div} \bar{A} = 0$ не явл. Лоренц. инвар.

5 1 пара ур-й Максвелла

$$\text{div} \bar{H} = 0$$

$$\frac{\partial F_{\mu\nu}}{\partial x^\nu} = 0$$

$$\text{rot} \bar{E} = -\frac{1}{c} \frac{\partial \bar{H}}{\partial t}$$

$$\tilde{F}^{\mu\nu} = \frac{1}{2} \varepsilon^{\mu\nu\alpha\beta} F_{\alpha\beta}$$

$$\frac{\partial \tilde{F}^{\mu\nu}}{\partial x^\mu} = \frac{\partial}{\partial x^\mu} \left(\frac{1}{2} \varepsilon^{\mu\nu\alpha\beta} \left(\frac{\partial A_\beta}{\partial x^\alpha} - \frac{\partial A_\alpha}{\partial x^\beta} \right) \right) = \frac{1}{2} \varepsilon^{\mu\nu\alpha\beta} \left(\frac{\partial^2 A_\beta}{\partial x^\mu \partial x^\alpha} - \frac{\partial^2 A_\alpha}{\partial x^\mu \partial x^\beta} \right) \stackrel{(*)}{=} 0$$

$$\tilde{F}^{\mu\nu} = \begin{pmatrix} 0 & -H_x & -H_y & -H_z \\ H_x & 0 & E_z & -E_y \\ H_y & -E_z & 0 & E_x \\ H_z & E_y & -E_x & 0 \end{pmatrix}$$

→ получим в точности пару ур-й Максвелла

$$\oint \bar{H} d\bar{S} = 0$$

$$\oint \bar{E} d\bar{l} = -\frac{1}{c} \int d\bar{S} \frac{\partial \bar{H}}{\partial t} = \frac{1}{c} \frac{\partial \varphi}{\partial t}$$

(*) пояснение

вертикаль или горизонт. с антисимм.

$$\varepsilon^{\mu\nu\alpha\beta} \frac{\partial^2}{\partial x^\mu \partial x^\alpha} = \frac{1}{2} \left(\varepsilon^{\mu\nu\alpha\beta} \left(\frac{\partial^2}{\partial x^\mu \partial x^\alpha} + \frac{\partial^2}{\partial x^\alpha \partial x^\mu} \right) \right) = 0$$

